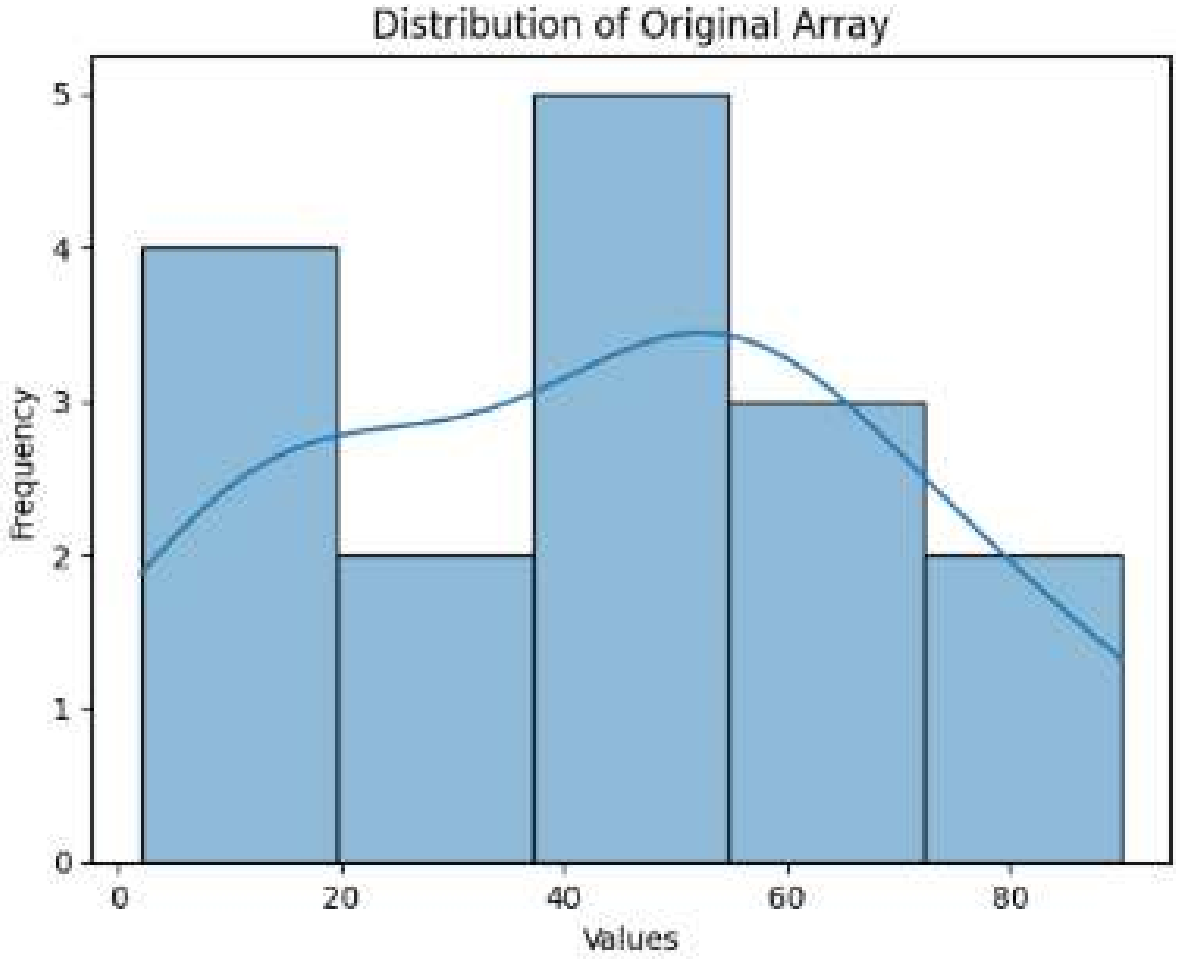
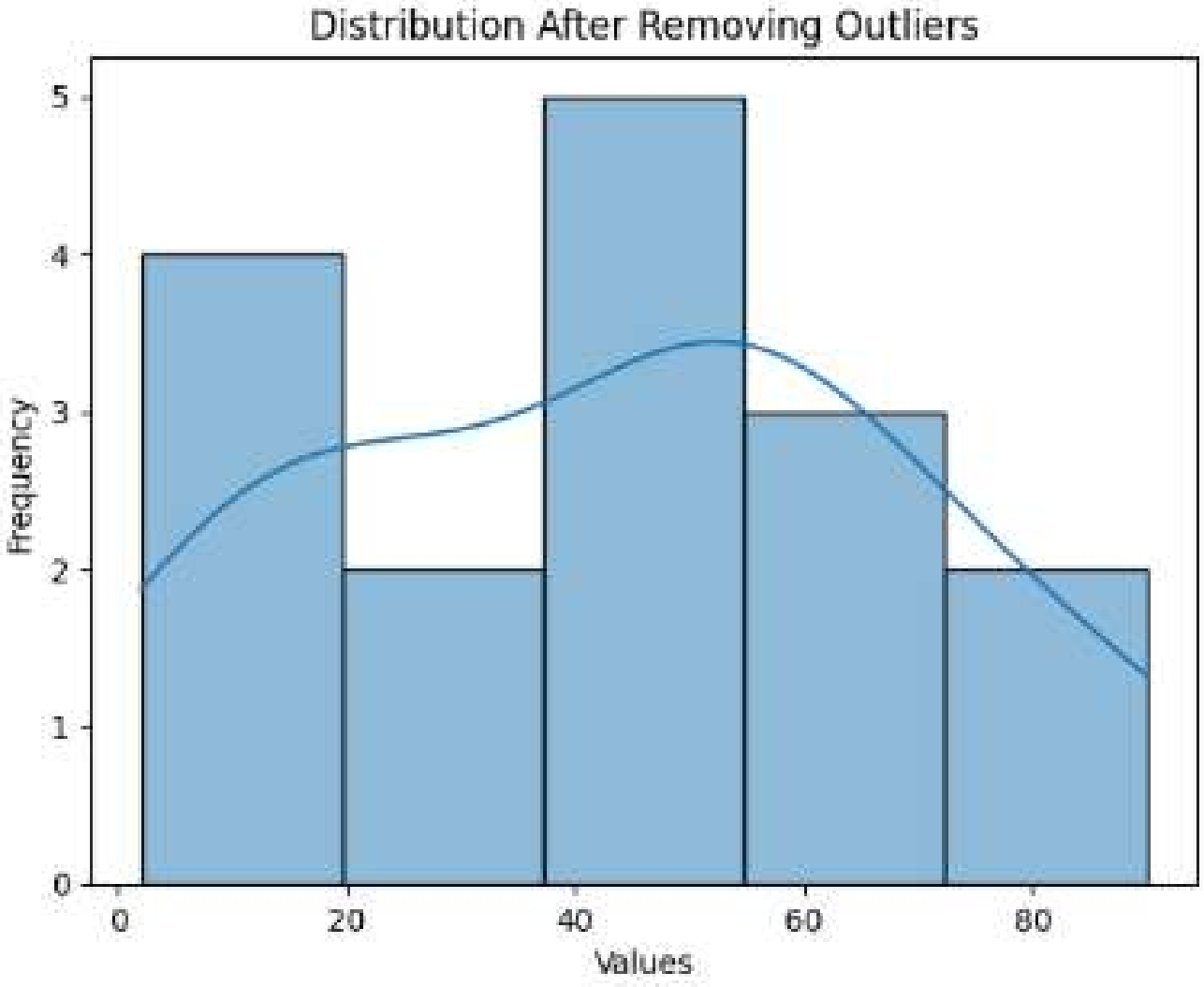
```
final_array = new_array[
    (new_array >= lr1) & (new_array <= ur1)
]

print("\nFinal Array:")
print(final_array)


# ----------------------------------------
# 9. Plot the final distribution
# ----------------------------------------

sns.histplot(final_array, kde=True)
plt.title("Final Array Distribution")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.show()
```

[6] ✓ 5.5s

```
Original Array:
[84  2 54 90 24 16 43  8 60 69 23 65 51 47 49 15]

Mean:
43.75

25th Percentile (Q1):
21.25

50th Percentile (Median):
48.0

75th Percentile (Q3):
61.25

100th Percentile (Maximum):
90.0

Original Array Outlier Boundaries:
Lower Range: -38.75
Upper Range: 121.25
```

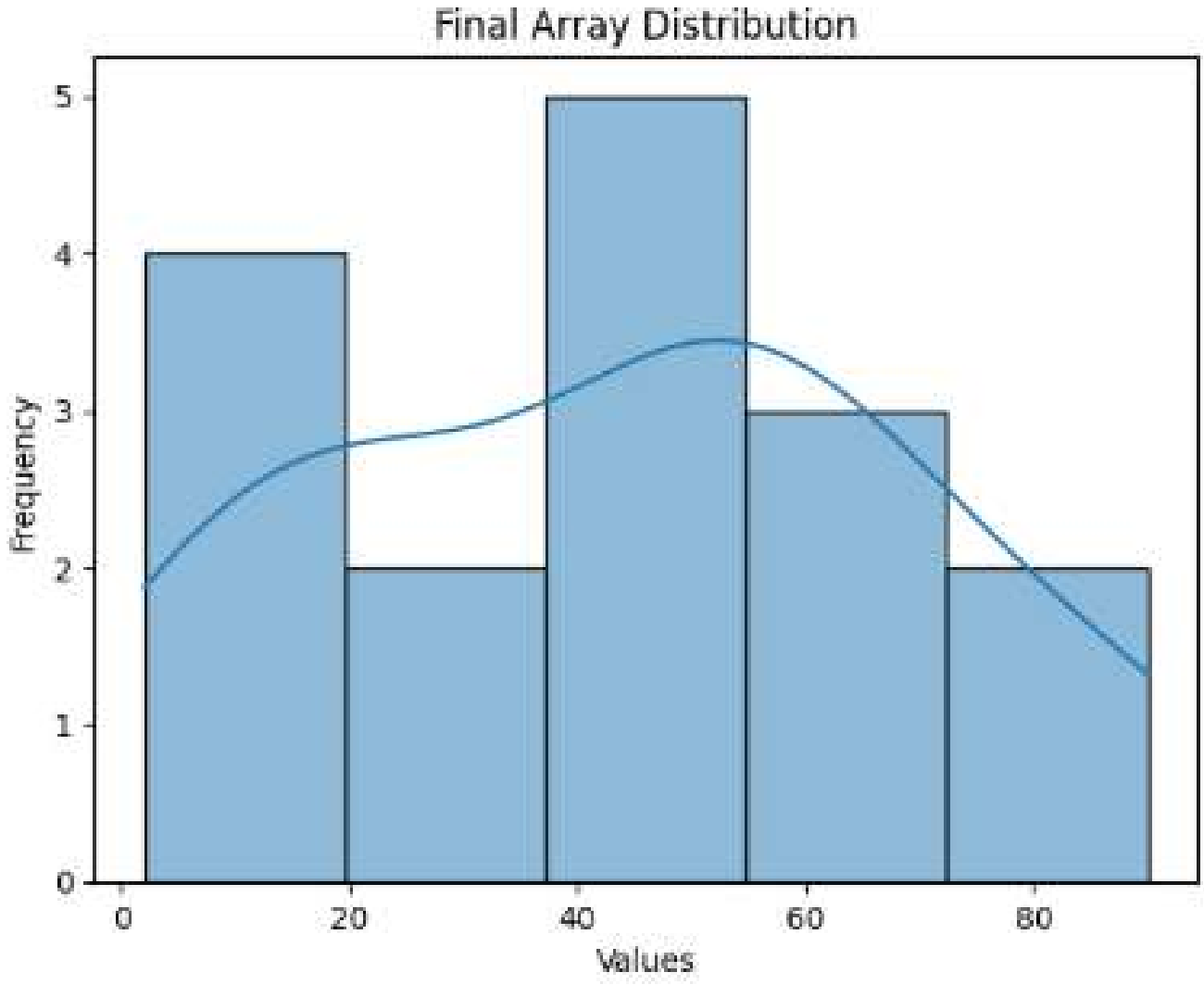

Distribution of Original Array
Frequency
Values
0
1
2
3
4
5
0
20
40
60
80

Array After Removing Outliers:
[84 2 54 90 24 16 43 8 60 69 23 65 51 47 49 15]



New Outlier Boundaries:
Lower Range: -38.75
Upper Range: 121.25

Final Array:
[84 2 54 90 24 16 43 8 60 69 23 65 51 47 49 15]

New Outlier Boundaries:
Lower Range: -38.75
Upper Range: 121.25

Final Array:
[84 2 54 90 24 16 43 8 60 69 23 65 51 47 49 15]